



LAN Switching: Switch Concepts

Network Switches

- A **Network Switch** is a component of computer **network** that connects two **network** segments and/or two **network** devices (**switches** or routers) together.
- Generally, **Switches** can also be referred to as **network** bridges with multiple ports (i.e. multiport bridges)
- Switches help to process and route packets at data link layer of the OSI reference model

Ethernet and LAN Switching



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- LAN operation requires a network **access method** as the basis for its operation
- **Ethernet** is the most common network access method in use today for local area networks
- Ethernet multi-access broadcast technology i.e. it is a shared technology where all users on a given LAN segment compete for the **same available bandwidth**
- It uses **CSMA/CD**
- Collisions impact on network performance
- **Layer 2** devices can improve performance

Ethernet and LAN Switching



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- With network growth bandwidth needs increased
 - Internet/intranet/email
 - Multimedia
 - Increasing use of enterprise servers
- Ethernet has developed to meet this challenge
 - 10Mbps, 100Mbps, 1000Mbps, 10Gbit
 - Coaxial, Twisted Pair, Fibre Optic, Wireless
 - Repeaters, hubs, bridges, switches, routers
- **NB:** You must understand the features of all these technologies to gain best performance in your network design!

Types of Ethernet Transmission



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Half-duplex

- Host checks medium for signal – if clear host transmits
- Only 1 host can transmit at a time
- Collisions – jam signal generated, back-off algorithm before retransmission
- 50-60% bandwidth available

Full duplex

- Host can transmit immediately
- 2 hosts can transmit simultaneously
- No collisions
- 100% bandwidth available
- Requires dedicated connection to a switch port

Types of Ethernet Transmission



- Benefits of using full-duplex:
 - i. Time is not wasted retransmitting frames because collisions do not occur
 - ii. The full bandwidth is available in both directions because the send and receive functions are separate
 - iii. Stations do not have to wait until other stations complete their transmissions because only one transmitter is used for each twisted pair

Limitations of Ethernet



1. Collisions which affect network performance
2. Distance – the coverage is within a short range (with older technology)
3. Broadcast storms
4. Security concerns

Growing Complexity of Networks



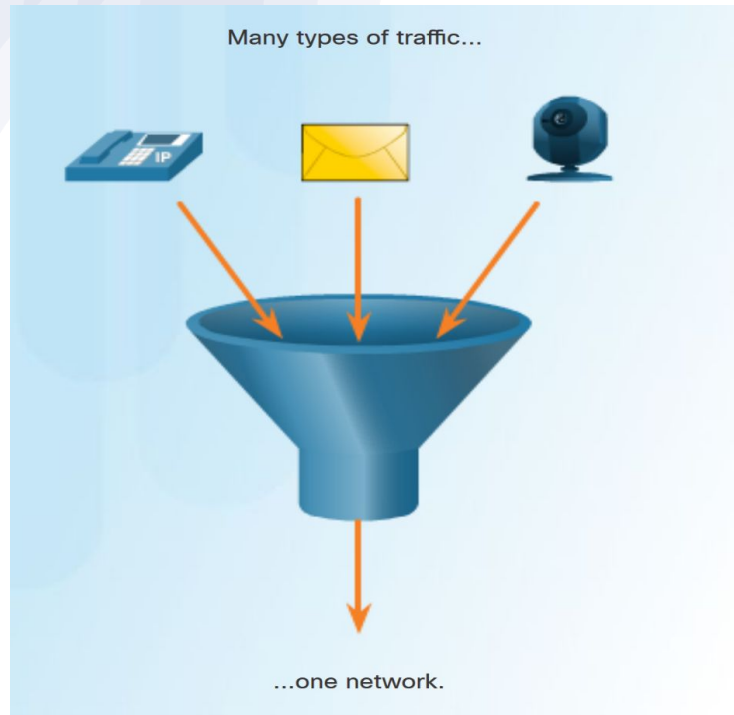
- Next-generation networks need to be secure, reliable, and highly available.
- They must support a globalized workforce.
- They must be able to integrate legacy devices.



LAN Segmentation



Elements of a Converged Network



- Converged network solutions integrate voice systems, IP phones, voice gateways, video support, and video conferencing.
- Primary benefit of the converged network - just one physical network to install and manage.

LAN Segmentation

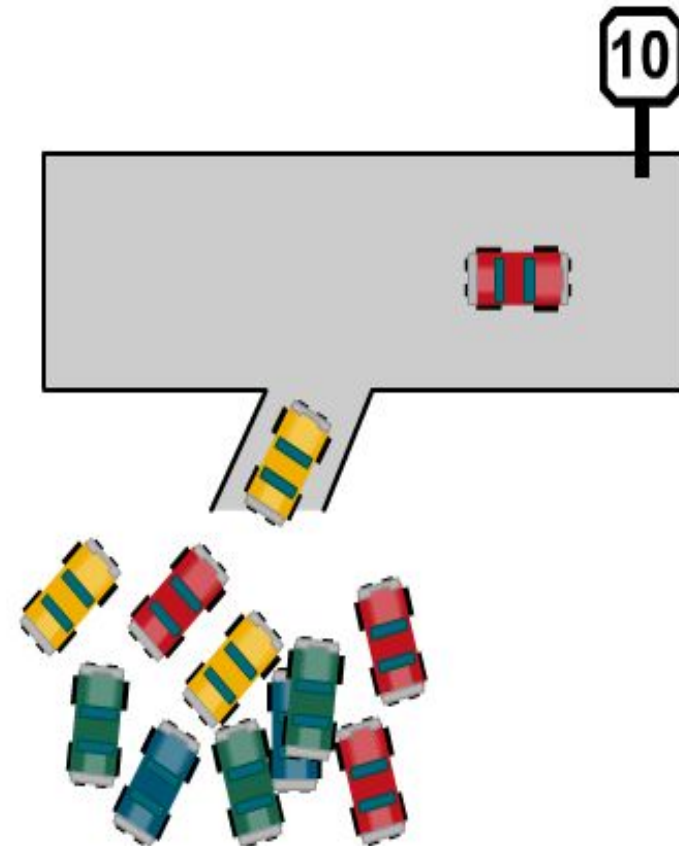


- You can improve the performance of your Ethernet network by reducing the number of stations per collision domain
- Typically, network administrators implement bridges, switches, or routers to segment the network and divide the collision domains

Segmenting with Hubs

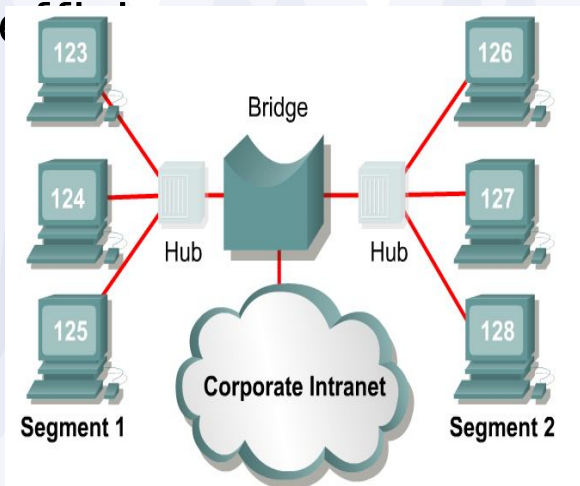


- Layer 1 devices
- Regenerate, retime, amplify signals
- **1 collision/bandwidth domain**
- Broadcasts propagated out of every port
- Only 1 device can transmit at a time
- Only 50-60% bandwidth available



Segmenting with Bridges

- Bridges are layer 2 devices
- They split a network into more collision/bandwidth domains
- Are capable of collecting and selectively passing data frames between two network segments
- They do this by learning the MAC address of all devices on each connected segment and use the information to build a bridge table
- They then forward or block traffic based on the bridge table
- This results in smaller collision domains and greater network



NB: Bridges do **NOT** restrict
broadcast traffic

- They add 10-30% latency

Segmenting with Bridges (cont.)



Points to remember about bridges:

1. Bridges reduce collisions on the LAN and filter traffic based on MAC addresses
2. A bridge does not reduce **broadcast** or **multicast** traffic
3. A bridge can extend the useful distance of the Ethernet LAN
4. The bandwidth for the new individual segments is increased
5. Bridges can be used to limit traffic for security purposes

Segmenting with Routers

- Routers operate at layer 3 of the OSI reference model
- Used to “route” traffic between two or more Layer 3 networks
- Routing decisions are made based on groups of network addresses, or classes, as opposed to individual Layer 2 MAC addresses
- They use **routing tables** to record the Layer 3 addresses of the networks that are directly connected to the local interfaces and network paths learned from neighboring routers
- They **do not** propagate broadcast traffic; thus, they reduce

Segmenting with Routers (cont.)



When you segment a LAN with routers, they will:

1. Decrease collisions by filtering traffic
2. Reduce broadcast and multicast traffic by blocking or selectively filtering packets
3. Support multiple paths and routes between them
4. Provide increased bandwidth for the newly created segments
5. Increase security by preventing packets between hosts on one side of the router from propagating to the other side of the router

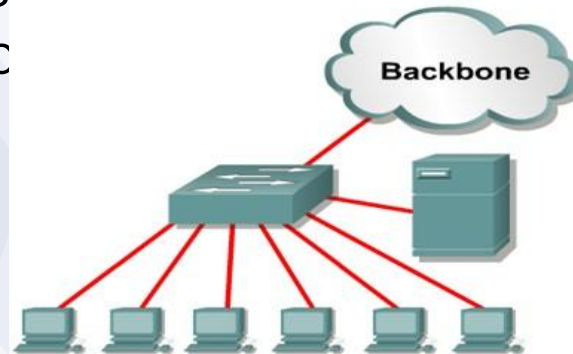
Segmenting with Routers (cont.)



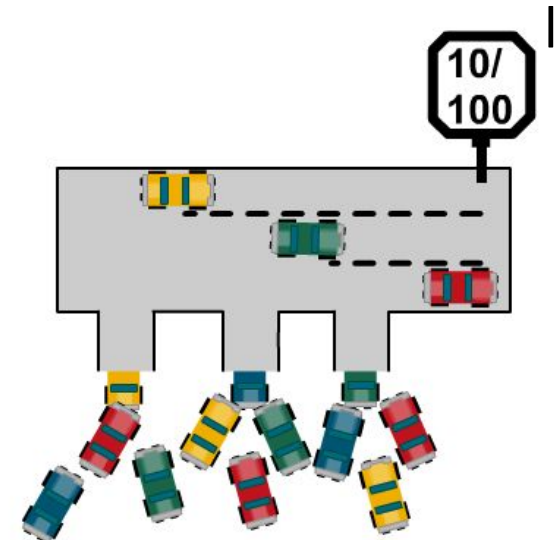
6. Increase the effective distance of the network by creating new collision domains
7. Provide layer 3 routing, packet fragmentation and reassembly, and traffic flow control
8. Provide communications between different technologies, such as Ethernet and Token Ring or Ethernet and Frame Relay
9. Have a higher latency than bridges, because routers have more to process; faster processors in the router can reduce some of this latency

Segmenting with Switches

- Layer 2 devices that learn MAC addresses of devices attached to each port and build a switching table. They use this switching table to selectively forward traffic
- They create a virtual circuit between two connected devices, establishing a dedicated communication path between two devices
- They **microsegment** the network by connecting each port to an individual workstation. Each switch port therefore has dedicated bandwidth (100% bandwidth is available)
- A switch is also able to facilitate multiple circuit connections



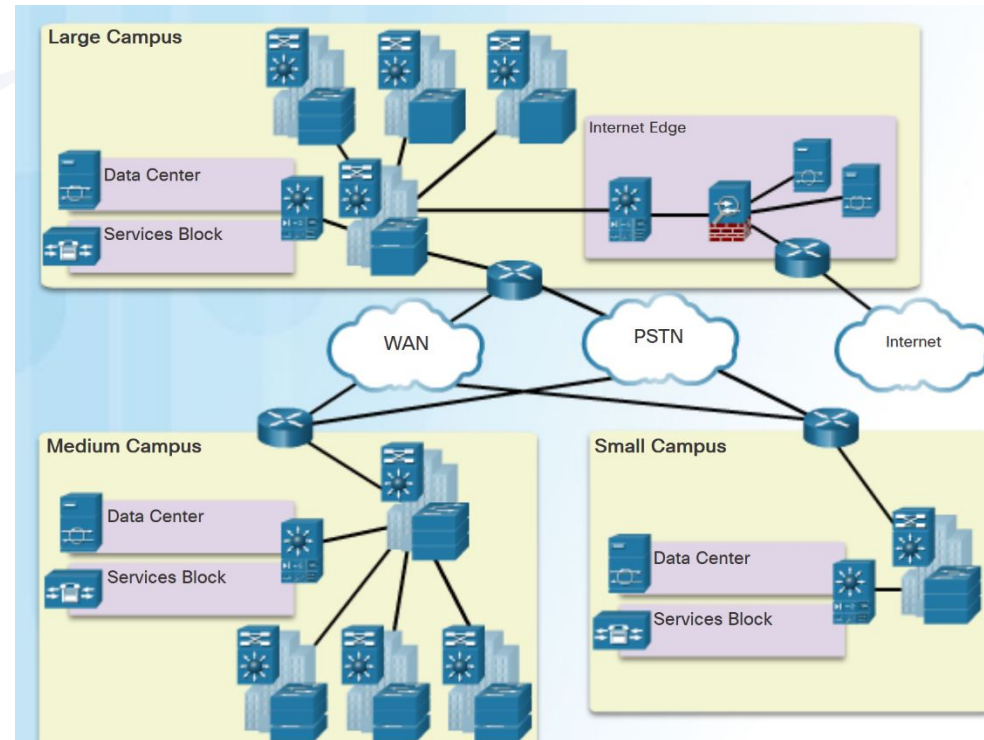
Each node has 10/100 Mbps



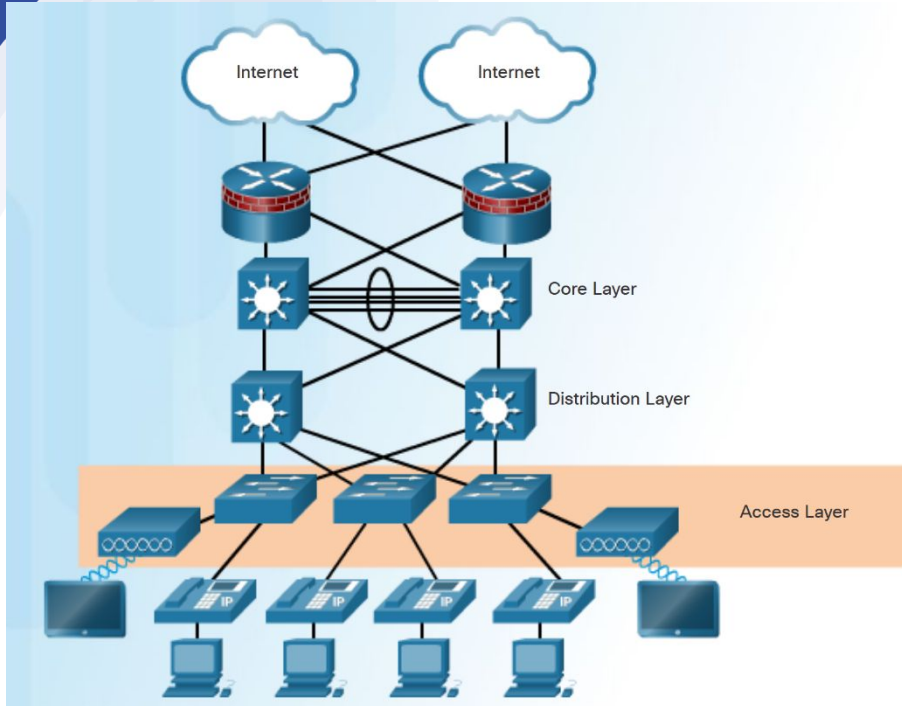


Cisco Borderless Networks

- The Cisco Borderless Network has the following features:
 - Allows organizations to connect anyone, anywhere, anytime, on any device; securely, reliably, and seamlessly.
 - Provides the framework to unify wired and wireless access, including policy, access control, and performance management across many different device types.
 - Provides network services, and user and endpoint services that are all managed by an integrated management solution.



Hierarchy in the Borderless Switched Network



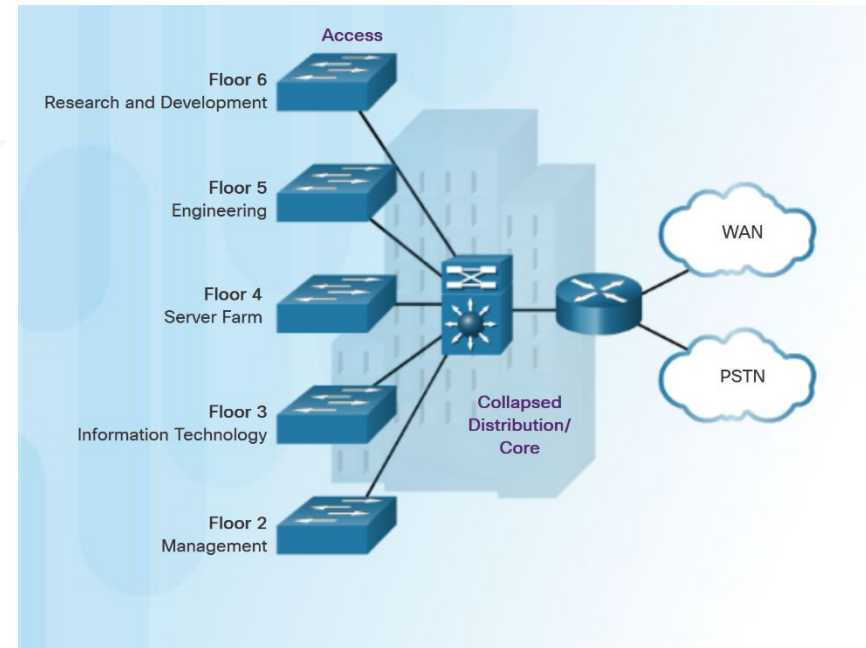
- Borderless switched network design guidelines are based on the following principles:
 - Hierarchical - Facilitates understanding the role of each device at every tier.
 - Modularity - Allows seamless network expansion and integrated services.
 - Resilience – Provides an always available network.
 - Flexibility - Allows intelligent traffic load sharing.
- The three tiers of the hierarchical model are Access, Distribution and Core layers.



LAN segmentation Layers

Access, Distribution, and Core Layers

- Access Layer – provides network access to the user.
- Distribution Layer - interfaces between the access layer and the core layer. Provides functions such as:
 - aggregating Layer 2 broadcast domains and Layer 3 routing boundaries.
 - providing intelligent switching, routing, and network access policy functions to access the rest of the network.
- Core Layer - is the network backbone. It provides fault isolation and high-speed backbone connectivity.

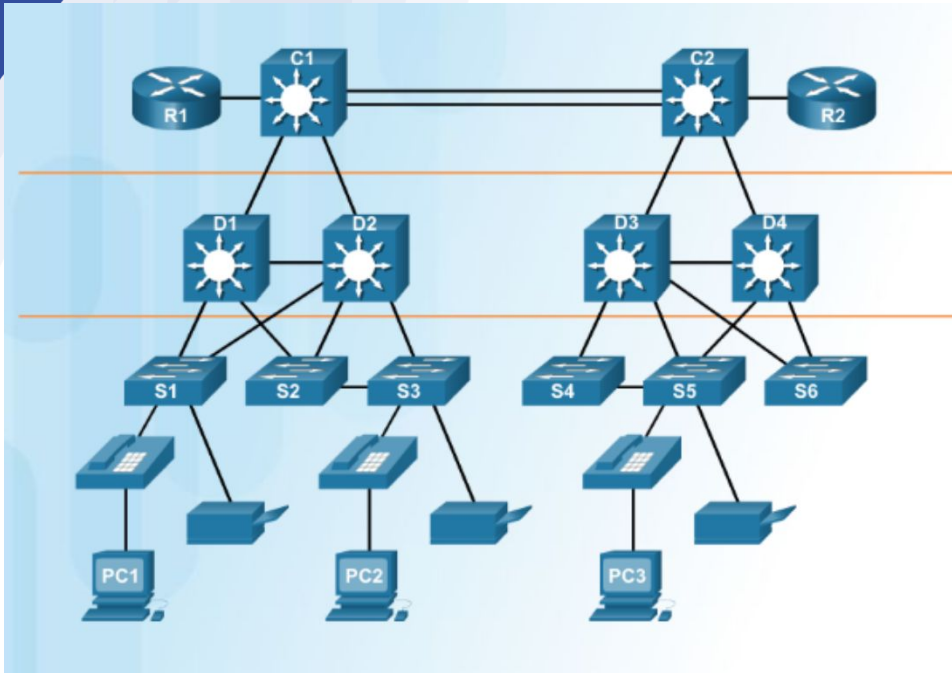


Smaller networks that do not need a separate distribution and core layer often use a two-tier campus or collapsed core network design.

LAN Segmentation

Role of Switches

- A hierarchical switched LAN allows more flexibility, traffic management, and additional features:
 - Quality of service
 - Additional security
 - Support for wireless networking and connectivity
 - Support for new technologies.



Switch Considerations: Form Factors



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Fixed Configuration



Stackable Configuration



Modular Configuration

- Considerations when selecting switches:
 - Cost
 - Port Density
 - Power
 - Reliability
 - Port Speed
 - Frame buffers
 - Scalability

Switch Operation

- When a switch starts up it sends a broadcast out of all ports to learn host MAC addresses
- When a frame is received for **an unknown destination** a broadcast is sent to discover its location
- Addresses are added to a switching table mapping them to the port on which they were learned
- When a frame is received for **a known destination** it is switched to the appropriate port

The Content Addressable Memory (CAM) Table

- CAM (Content Addressable Memory) is special type of memory used by Cisco switches. In the case of ordinary RAM the IOS uses a memory address to get the data stored at this memory location, while with CAM the IOS does the inverse; *It uses the data and the CAM returns the address where the data is stored.*
- This is done in a single CPU cycle vs. the traditional programming of searching through a table, which will cost many CPU cycles. Thus the CAM is considered to be faster than the RAM since the CAM searches the entire memory in one operation.
- The CAM table is present in all switches for layer 2 switching allowing the switches to facilitate communications between *connected nodes at high speed and in full-duplex regardless of how many devices are connected.*
- The CAM table is used by the *Ethernet switch logic* to store information such as *MAC addresses available on physical ports with their associated VLAN Parameters).*

Switch Architecture

Content Addressable Memory (CAM) Table

MAC Address	Port	VLAN
0000.0000.1111	1	1
0000.0000.2222	2	1
0000.0000.6666	6	1
0000.0000.5555	5	1
0000.0000.3333	3	20
0000.0000.4444	4	1

Frame for
0000.0000.5555



Broadcast
Frame



- A switch receives incoming link-layer frames, *filters* and *forwards* them onto outgoing links. Switch filtering and forwarding are done with a *CAM table*, also variably referred to as *switch*, *switching table* or *MAC table*.

Content Addressable Memory (CAM) Table

- **So how exactly does the CAM Table work?**
- A CAM tables provide only two results: 0 (true) or 1 (false). CAM is most useful for building tables that search on exact matches such as MAC address tables. The CAM table is the primary table used to make Layer 2 forwarding decisions. In the case of Layer 2 switching tables, *the switch must find an exact match to a destination MAC address or the switch floods the packet out all ports in the VLAN.*
- The table is built by *recording the source address and inbound port of all frames. As frames arrive on switch ports, the source MAC addresses are learned and recorded in the CAM table. The port of arrival and the VLAN are both recorded in the table, along with a timestamp.*
- If a MAC address learned on one switch port has moved to a different port, the MAC address and timestamp are recorded for the most recent arrival port and the previous entry deleted. If a MAC address is already present in the table for the correct arrival port only its timestamp is updated

Remember the ARP table?*

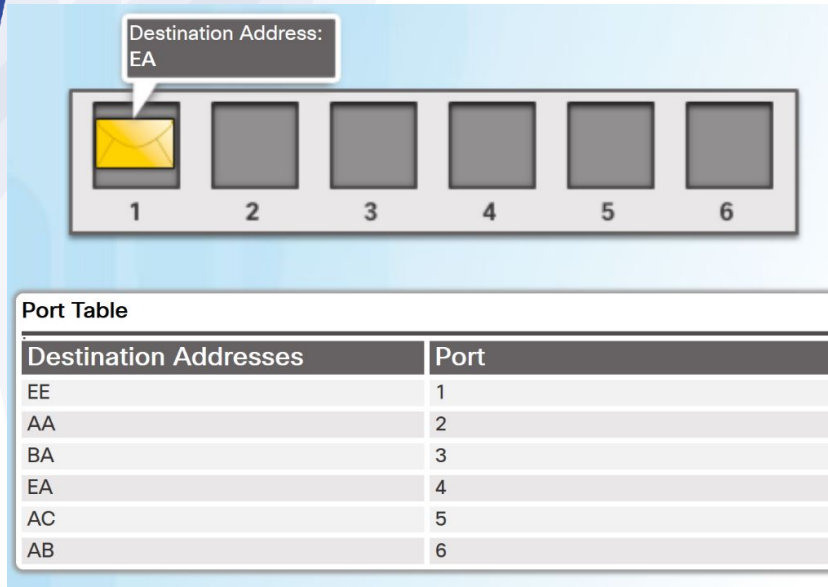
- The ARP table uses the ARP protocol to map MAC Addresses to IP addresses. If no ARP entry exists, an ARP broadcast is sent out, and the table is updated with the response.

```
C:\Users\lgachoki>arp -a
Interface: 192.168.100.15 --- 0xc
    Internet Address      Physical Address      Type
    192.168.100.1         9c-71-3a-92-ea-a0     dynamic
    192.168.100.3         64-1c-b0-00-95-19     dynamic
    192.168.100.255       ff-ff-ff-ff-ff-ff     static
```

Difference between the CAM table and ARP table?

- There are varying views and definitions of the terms CAM, MAC, ARP tables with some sources using them interchangeably. But basically...
- A *CAM table* maps *Ethernet hardware addresses* to *switch ports*. There is only one CAM table found in the switch.
- An *ARP table* maps *IP addresses* to *Ethernet hardware addresses*. Each host (TCP/IP client) connected to the network has its own ARP Table.

Switch Operation



- A LAN switch makes decisions based on two criteria:
 - Ingress port** - where a frame enters the device
 - Destination address**

- A LAN switch maintains a table that it uses to determine how to forward traffic
- In the diagram, If a message enters switch port 1 with a destination address of EA, then the switch forwards the traffic out port 4
- Layer 2 Ethernet switches forward frames based on the destination MAC address.

Switch CAM Table

- Switches contain RAM – known as CAM “Content Addressable Memory” which stores the MAC address table
- CAM is also used as frame buffer to queue frames in asymmetric switching – switch ports operating at different speeds e.g. 10 and 100 Mbps
- The table matches switch port with MAC address of attached device
- It is built by inspecting source MAC address of incoming frames, just like the bridge

Switch CAM Table



- Destination MAC address is checked against table, frame sent through correct port
- If the destination MAC address and port mapping is not in table, the frame flooded
 - NB: Broadcasts are flooded

Switches and Collision Domains

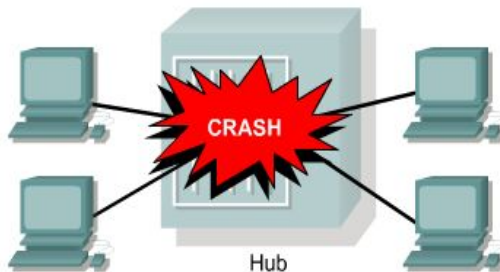


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- The network area where frames originate and collide or a network segment where devices must compete to communicate is called the **collision domain**. The network segments *that share the same bandwidth between devices* are in the same *collision domains*.



- "I could have walked to Finance by now."
- "I knew I should have stayed home."
- "File transfers take forever."
- "I'm waiting all the time."



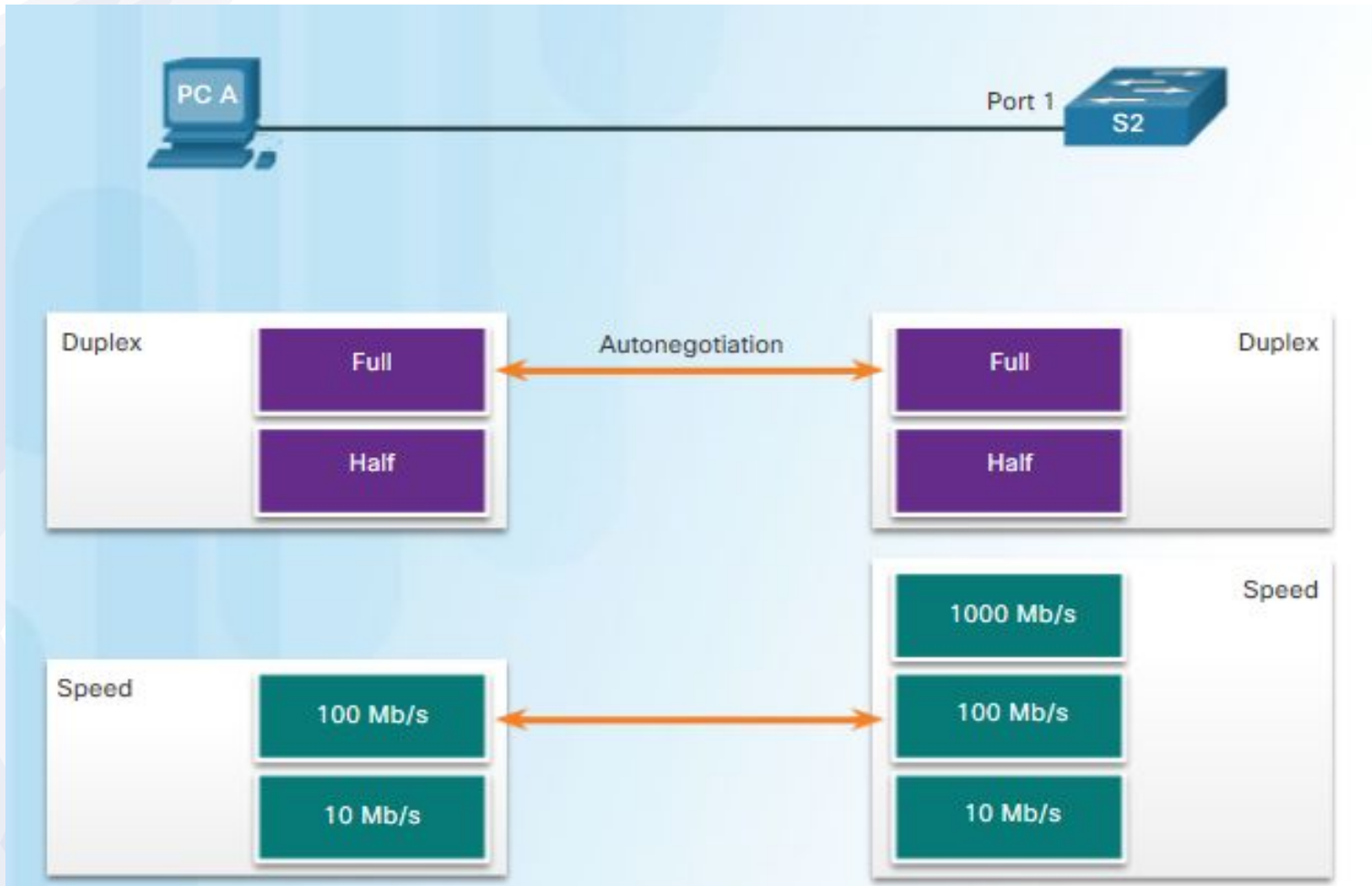
- All **shared media environments** are collision domains

- Sluggish network response
- Increasing user complaints

Switch Domains: Collision Domains

- In hub-based Ethernet segments, network devices *compete for the medium*, because devices must take turns when transmitting.
- *When two or more devices within that the same collision domain try to communicate at the same time, a collision will occur.*
- If an Ethernet switch port is operating in half duplex, each segment is in its own *collision domain*. However, Ethernet switch ports operating in full duplex eliminate collisions; therefore, there is no collision domain.
- By default, Ethernet switch ports will autonegotiate full duplex when the adjacent device can also operate in full duplex. If the switch port is connected to a device operating in half-duplex, such as a legacy hub, then the switch port will operate in half duplex. In the case of half duplex, the switch

Switch Domains: Collision Domains



full-duplex is chosen if both devices have the capability along with their highest common bandwidth.



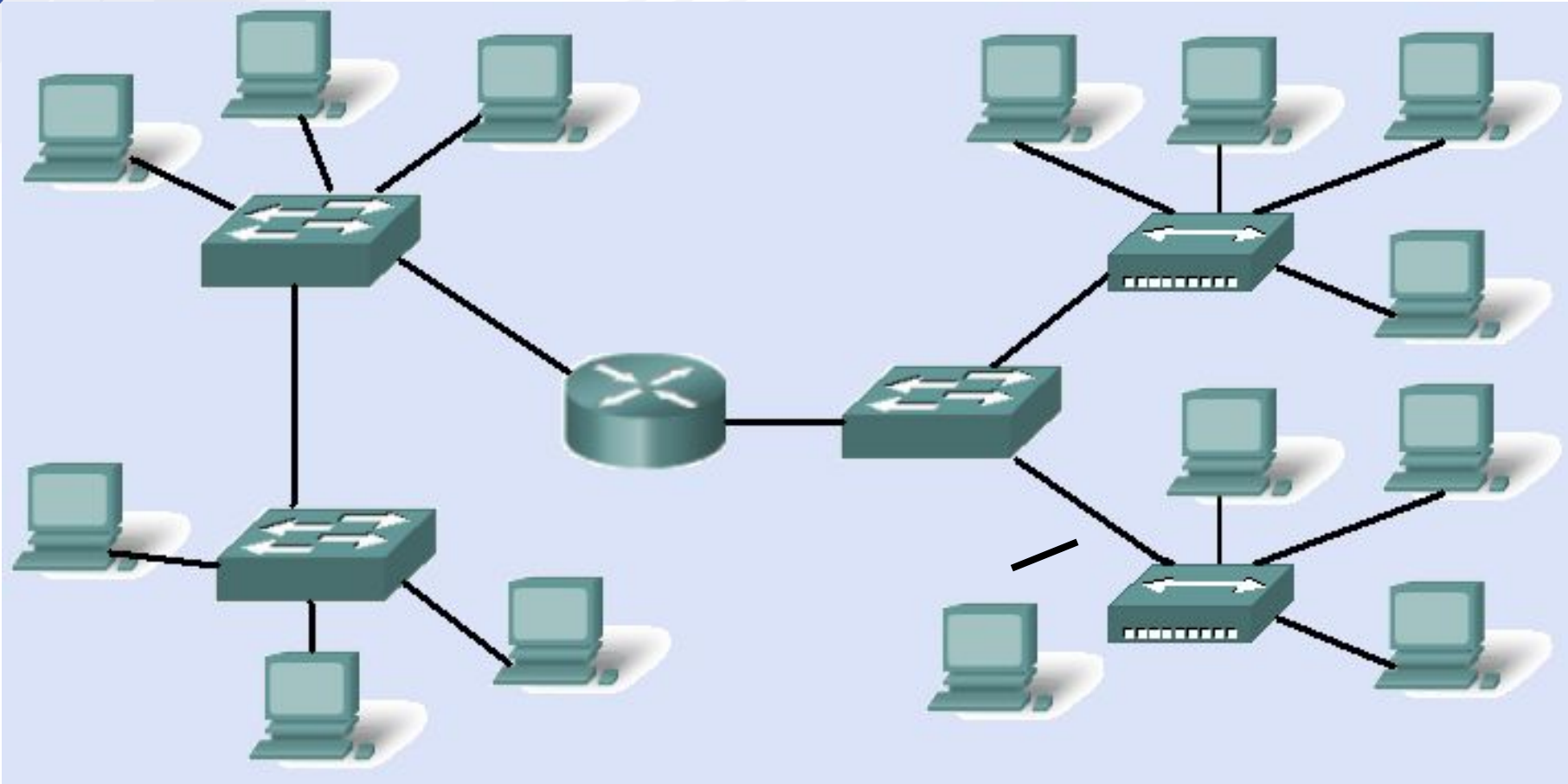
- Examples of Collision Domain Environments
 1. All ports of a hub belong to the same collision domain
 2. Every port of a switch is a collision domain on its own (*if configured to operate in full-duplex mode*)
 3. Every port of a router is a collision domain on its own
- NB:
 - Switches break the segment into smaller collision domains, easing device competition

Switches and Collision Domains

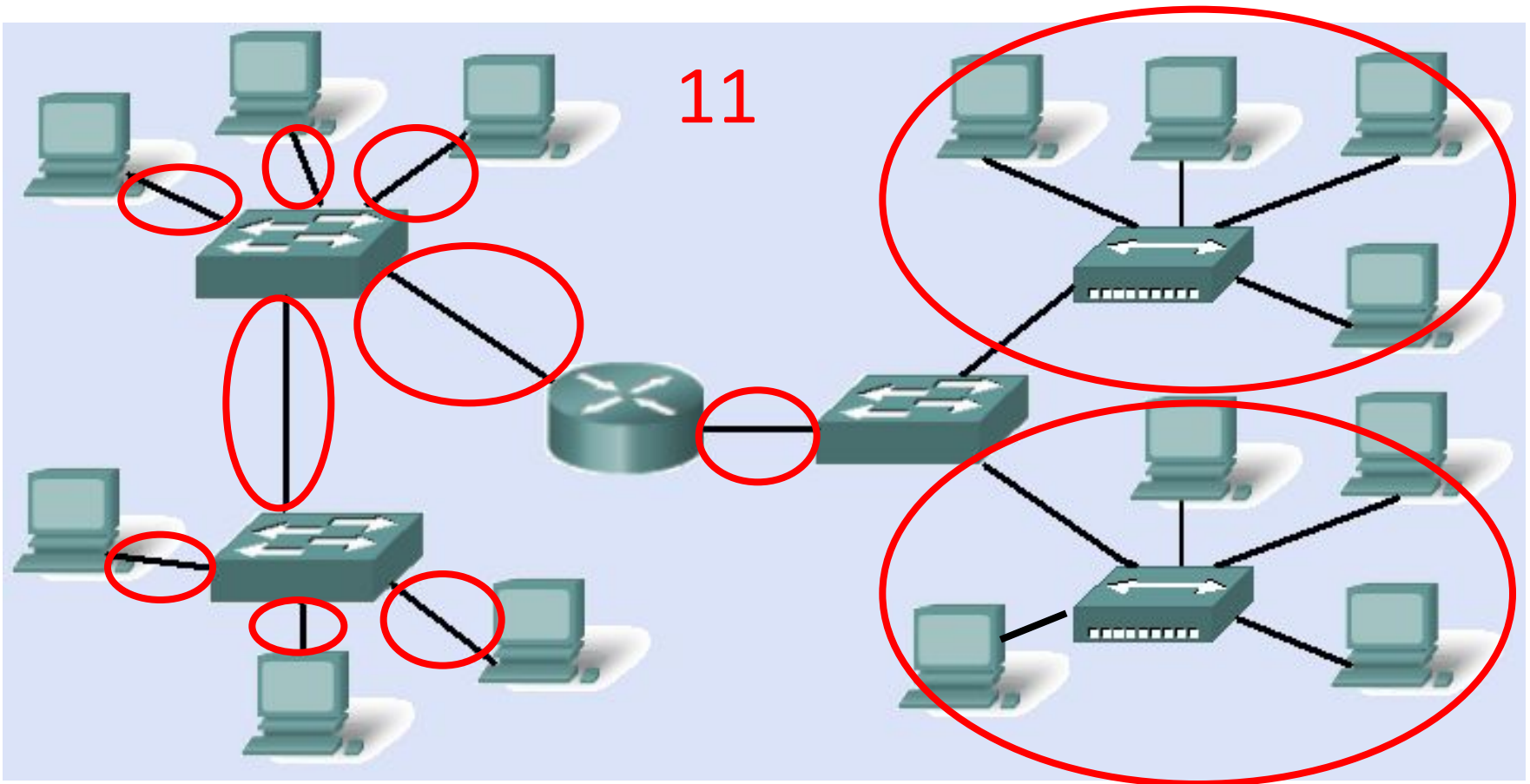


- Collisions reduce throughput
 - The more devices the more collisions
 - Using a hub there may be 60% of bandwidth available
 - Using a switch (+ full duplex transmission) dedicated link each way there's 100% bandwidth in each direction
- **NB:** The full duplex link above is regarded as an individual collision domain if you are asked to count them.

How many collision domains?



How many collision domains?



Switches and Broadcast Domains



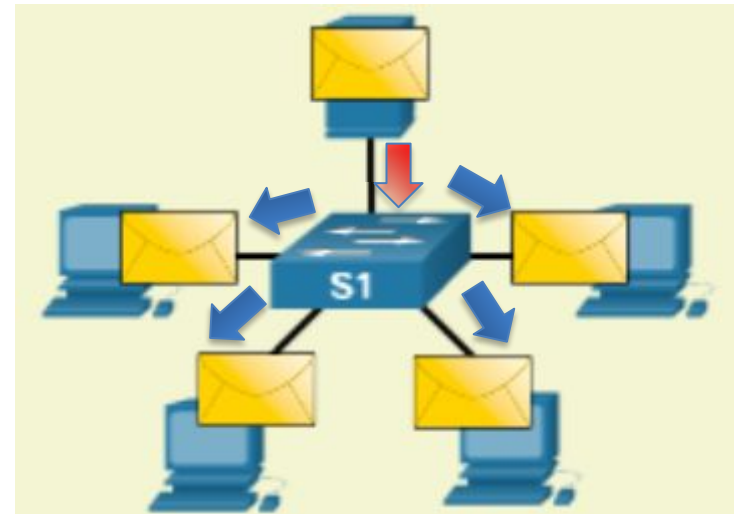
- Broadcasting is when one transmitter tries to reach all the receivers in the network i.e. the sending station sends out one message, and all nodes on that segment receive the message
- A **broadcast domain** is the extend of the network where a broadcast frame can be 'heard' or propagated
- Layer 2 switches flood broadcasts frames to all ports(when destination address does not exist); therefore, switches do not break broadcast domains. All ports of a switch, with its default configuration, belong to the same broadcast domain
- Devices linked by switches are in the same broadcast domain (if there are no VLANs configured)

Switches and Broadcast Domains

- If two or more switches are connected, broadcasts are forwarded to all ports of all switches, except for the port that originally received the broadcast
 - Destination MAC address for broadcast is all 1s, that is FF:FF:FF:FF:FF:FF
 - A layer 3 device (e.g. a router) splits up broadcast domains, does not forward broadcasts
- NB:** Layer 2 switches configured with VLANs break broadcast domains into smaller ones
- Too many broadcasts can cause **network congestion**.

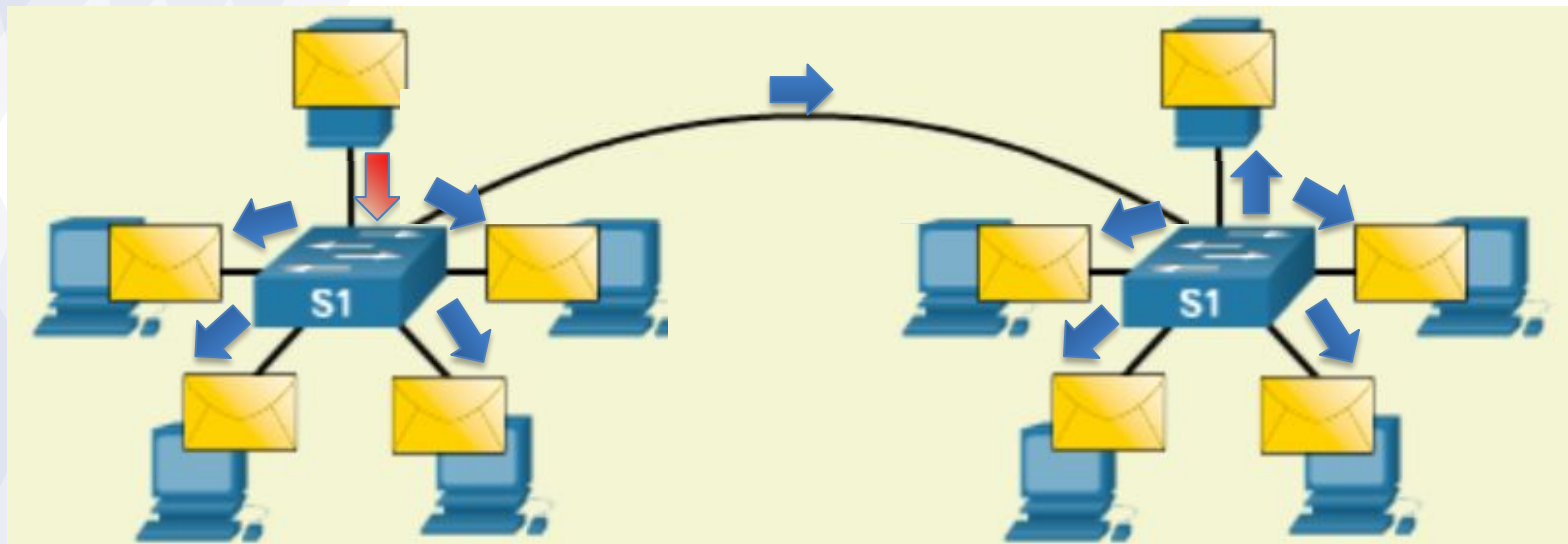
Switch Domains: Broadcast Domains

- A collection of interconnected switches forms a *single broadcast domain*. Only a network layer device, such as a router, can divide a Layer 2 broadcast domain. *Routers are used to segment broadcast domains, but will also segment a collision domain.*
- When a device sends a Layer 2 broadcast, the destination MAC address in the frame is set to all binary ones. The Layer 2 broadcast domain is referred to as the MAC broadcast domain. The MAC broadcast domain consists of all devices on the LAN that receive broadcast frames from a host.
- When a switch receives a broadcast frame, it forwards the frame out each of its ports, except the ingress port where the broadcast frame was received. Each device connected to the switch receives a copy of the broadcast frame and processes it.



Switch Domains: Broadcast Domains

- When two switches are connected together, the broadcast domain is increased, as seen in the second half of the animation. In this case, a broadcast frame is forwarded to all connected ports on switch S1. Switch S1 is connected to switch S2. The frame is then also propagated to all devices connected to switch S2



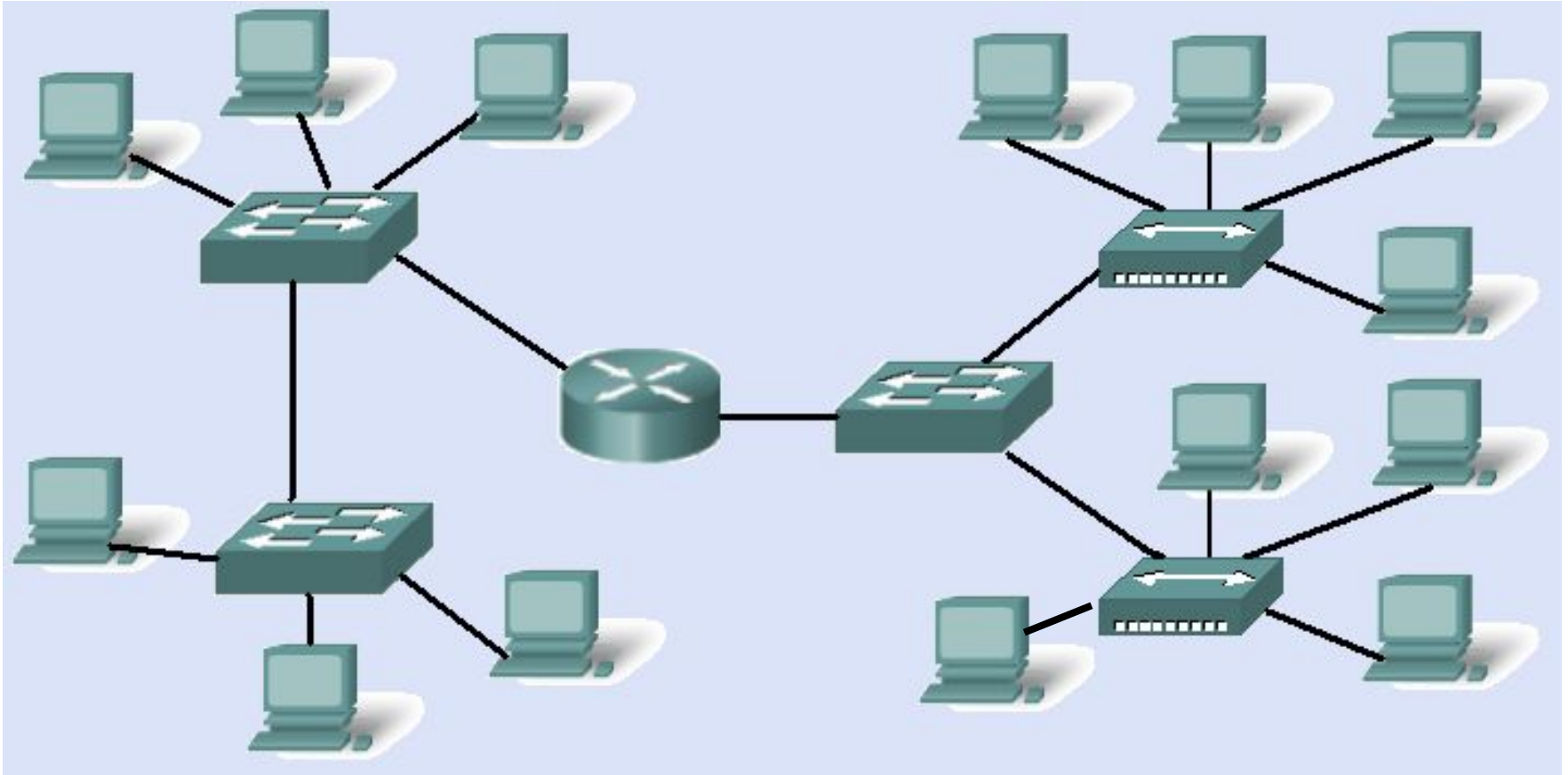
Pros:

- Broadcasts are sometimes necessary for initially locating other devices and network services.
- Fast and reliable communication for offices in different locations.

Cons:

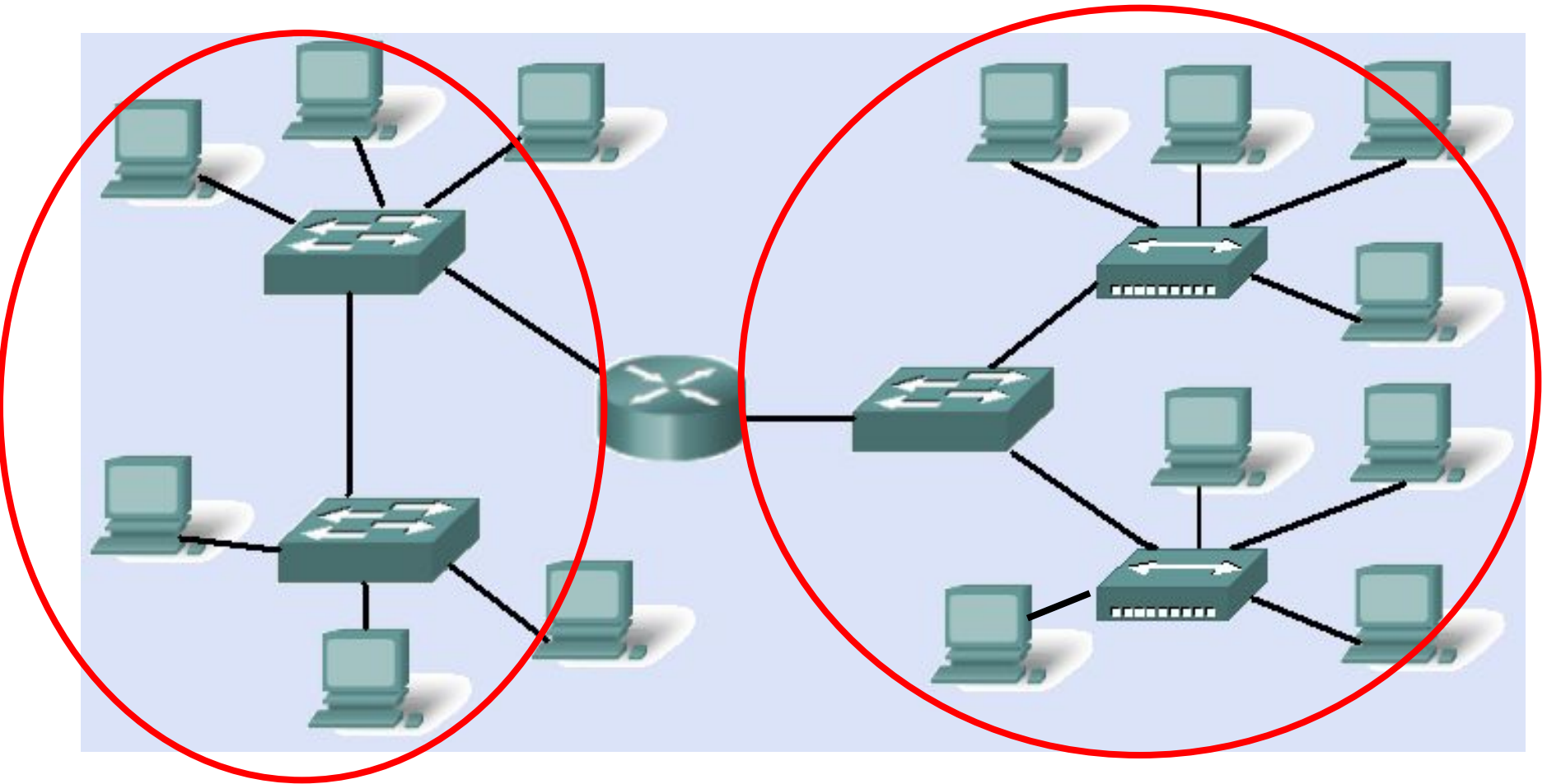
- They reduce network efficiency as network bandwidth is used to propagate the broadcast traffic.
- Too many broadcasts and a heavy traffic load on a network can result in congestion, which slows down network performance.
- E.g. tendency to drop Web data signals after reaching network router interface borders.

How many broadcast domains?

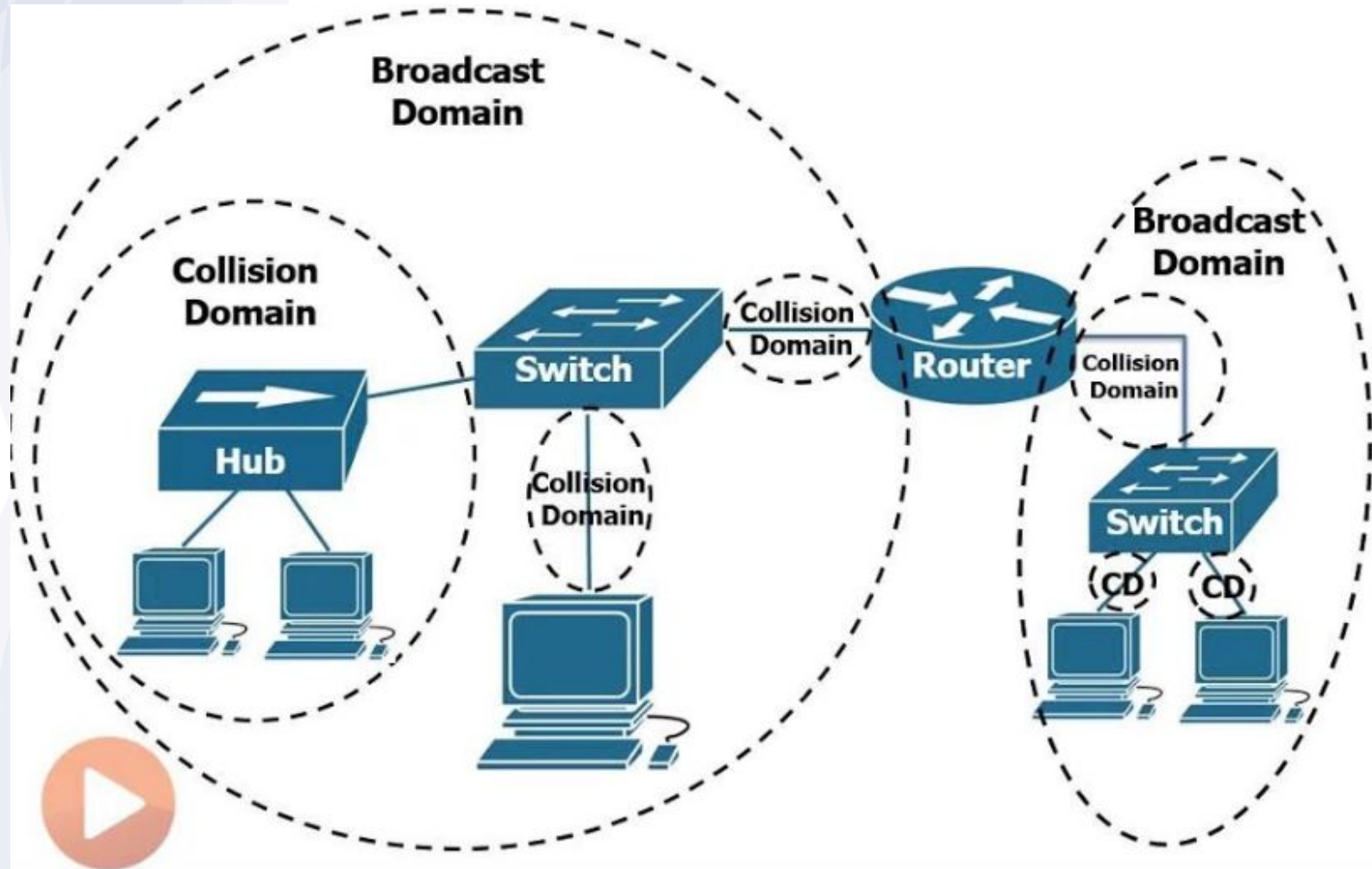


No VLANs

How many broadcast domains?



Broadcast and Collision Domains



Switch Domains: Alleviating Network Congestion

- LAN switches have special characteristics that make them effective at alleviating network congestion. By default, interconnected switch ports attempt to establish a link in full duplex, therefore eliminating collision domains.
- Each full duplex port of the switch provides the full bandwidth to the device or devices that are connected to that port.
- A full-duplex connection can carry transmitted and received signals at the same time. Full-duplex connections have dramatically increased LAN network performance, and are required for 1 Gb/s Ethernet speeds and higher.
- Switches interconnect LAN segments, use a table of MAC addresses to determine the segment to which the frame is to be sent, and can lessen or eliminate

Important characteristics of switches for alleviating Network Congestion

- *High port density* - Switches have high-port densities: 24- and 48-port switches are often just a single rack unit and operate at speeds of 100 Mb/s, 1 Gb/s, and 10 Gb/s. Large enterprise switches may support many hundreds of ports.
- *Large frame buffers* - The ability to store more received frames before having to start dropping them is useful, particularly when there may be congested ports to servers or other parts of the network.
- *Port speed* - Depending on the cost of a switch, it may be possible to support a mixture of speeds. Ports of 100 Mb/s, and 1 or 10 Gb/s are common (100 Gb/s is also possible).

Important characteristics of switches for alleviating Network Congestion

- *Fast internal switching* - Having fast internal forwarding capabilities allows high performance. The method that is used may be a fast internal bus or shared memory, which affects the overall performance of the switch.
- *Low per-port cost* - Switches provide high-port density at a lower cost.
- Facilitating the segmentation of a LAN into separate collision domains: Use of Switches and Bridges and avoiding hubs
- Providing full-duplex communication between devices:
Select devices that support full duplex communication



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